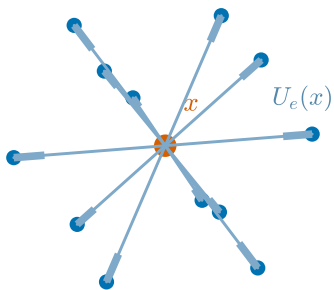
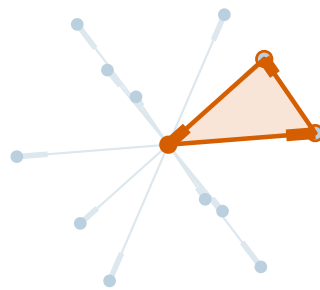


(a) link variables on $D_3 = \text{fcc}$: 12 per site



in $d=4$: 24 links, 12 independent directions

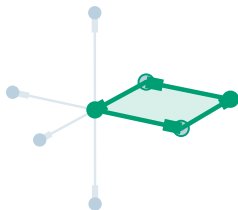
(b) the minimal loop is a triangle



$$\text{Tr } U_{\triangle} = \text{Tr} [U_{e_1} U_{e_2} U_{e_3}]$$

96 triangles through each site of D_4

(c) hypercubic: bipartite, so no triangles



the minimal loop is the 4-link plaquette

$\text{Tr } U_P$, 24 through each site of $\mathbb{Z}^4$

(d) loops through a site

